

Abhinav Gupta

abhinavrgupta98@gmail.com • +91-8209586719 • [linkedin.com/in/abhinav-gupta01](https://www.linkedin.com/in/abhinav-gupta01) • github.com/rgabhi

M.Tech. CS student at IIT Delhi with **3+ years** of industry experience at Fractal.ai and American Express, having strong foundations in **systems programming, algorithms** and distributed data. Seeking SDE internship roles.

EDUCATION

IIT Delhi

M.Tech. Computer Science
Jul 2025 – Present

IIT Guwahati

B.Tech. | CGPA: **8.48/10**
Jul 2017 – Jun 2021

SKILLS

Languages

Python C C++ SQL Java

CS Fundamentals

DSA System Design OOP

OS DBMS

Frameworks & Libraries

Spring TensorFlow Keras

OpenCV NumPy Pandas

Big Data & Cloud

PySpark Hadoop Azure ADLS

Databricks

Tools

Git Linux REST Tableau

Agile Vim

COURSEWORK

- Data Structures & Algorithms
- Systems Concepts & Prog.
- Database Management Systems
- Computer Vision
- Artificial Intelligence
- Graph Theory & Linear Algebra

ACHIEVEMENTS

800+ DSA Problems

Arrays, Graphs, DP, Trees on
LeetCode & Scaler.

Star Award '22 & '23

Best performance & client
appreciation, Fractal.ai

Samsung Fellowship 2021

Merit scholarship; in top 10% of
batch, IITG

JEE Advanced 2017

All India Rank in top 3% among 2.2
lakh candidates

Hackathon — 1st Place

CodeAsylums Web-Dev Hackathon

EXPERIENCE

American Express

Software Engineer — Data Platform

Gurgaon, India | Dec 2024 – Jun 2025

Tech Stack: Python, SQL, PySpark, Hadoop, GCP

- Architected a centralized data repository to unify global prospect data; implemented fuzzy-matching algorithms (Monge-Elkan, Jaro-Winkler) for high-accuracy entity resolution across millions of records.
- Designed and automated end-to-end campaign file generation and suppression pipelines, enforcing risk and marketing compliance and eliminating manual error-prone handoffs.

Fractal.ai

Data Engineer

Bengaluru, India | Aug 2021 – Dec 2024

Tech Stack: Python, PySpark, SQL, Azure, Databricks

- Built a high-throughput ETL pipeline processing **10M+ records**; achieved **75% runtime reduction** via concurrent processing, dataframe caching, and partition-aware execution strategies.
- Reduced dashboard query latency from **minutes to sub-seconds** by redesigning SQL query plans, introducing optimized indexing, and restructuring Azure ADLS schemas for *Polaris*, a portfolio optimization platform.
- Engineered a custom global product ranking algorithm to evaluate SKU performance at scale, enabling data-backed client decisions; recognised with the **Star Award** in 2022 & 2023.

PROJECTS

Mini UNIX Shell & Bytecode VM | C, Linux Systems API

[GitHub]

- Developed a UNIX command-line interpreter with a custom tokenizer, implementing Inter-Process Communication (IPC) via pipes and file descriptors (`dup2`) for I/O redirection.
- Built a stack-based Bytecode Virtual Machine (BVM) and assembler in C, featuring a custom garbage collection system for memory management.

Linux Process Debugger | C++, ptrace, Systems Programming

[GitHub]

- Built a Linux-based debugger in C++ using the `ptrace` system call to support breakpoints management, register inspection, and single-step execution flow.

Augmented Reality Tag Detection | Python, Computer Vision

[GitHub]

- Built an augmented reality application to detect and overlay 3D objects on AR tags to simulate real-world spatial tracking.
- Implemented contour detection, homography computation, and perspective warping entirely from scratch, strictly avoiding OpenCV functions for core logic.

AI Game Agent — Stones & Rivers | C++, Algorithms

[GitHub]

- Implemented a high-performance Minimax agent with Alpha-Beta pruning and transposition tables, maximizing search depth under real-time constraints using Manhattan distance territory control.

ANN-based Protein Toxicity Predictor | Python, TensorFlow, Keras

- Built an end-to-end data ingestion and feature engineering pipeline using UniProtKB data (~42K proteins) for drug design.
- Designed and trained a multi-layer feed-forward ANN with Adam optimizer, achieving ~93% validation accuracy evaluated via F1-score and confusion matrix.

Contactless Fingerprint Recognition | Python, Computer Vision

[GitHub]

- Developed a contactless fingerprint recognition system by implementing a robust feature extraction, image enhancement, and minutiae matching pipeline.